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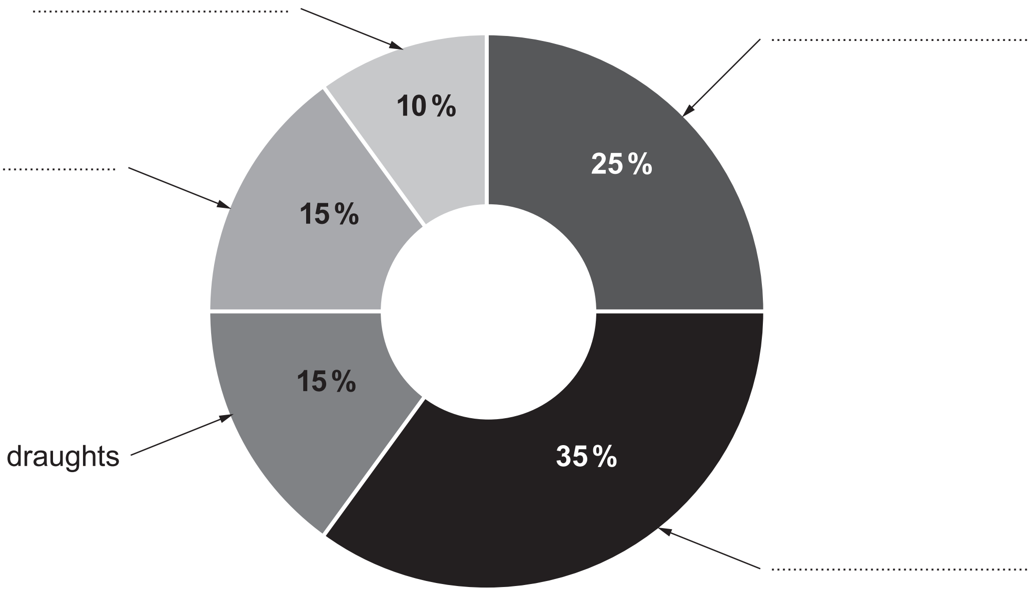
7. The Welsh Government Warm Homes Scheme, called Nest, aims to make Welsh homes warmer and more energy efficient places to live. Nest is accessible to all homeowners in Wales and provides advice on saving energy.

(a) Draughts, floors, windows, walls and the roof are the five ways energy is lost from a heated house. The pie chart shows the percentages of energy loss by each of the five ways.

Use the information below to **complete the labelling** on the pie chart. [3]
One label has been completed already.

- The percentage energy loss through the walls is greatest.
- Windows and draughts have equal percentage energy losses.
- Windows and floor percentage losses add up to equal the percentage loss through the roof.



(b) To reduce energy loss through the roof the Nest Scheme suggests the installation of fibre-glass insulation in the loft.

(i) Explain how fibre-glass reduces energy loss in the loft by convection. [2]

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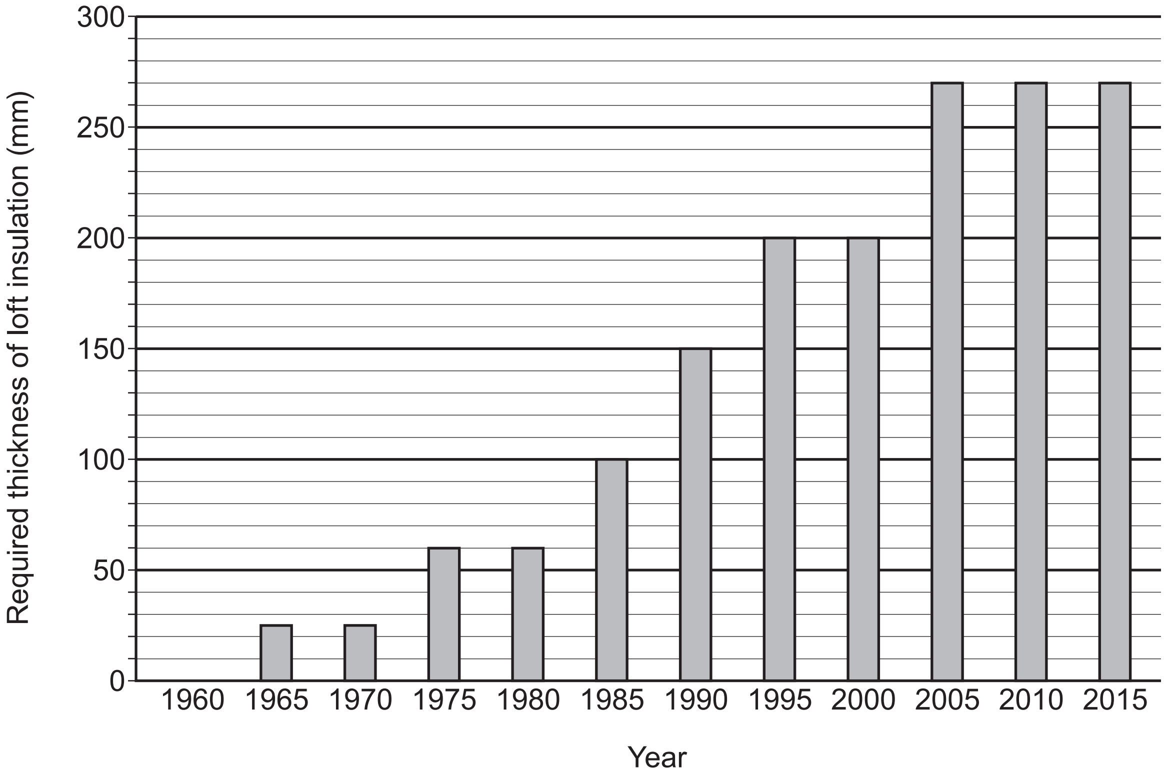
(ii) There are strict regulations about the minimum thickness of insulation that must be installed in the loft of a new house. The graph shows how the required thickness of loft insulation has changed since 1960.



Use information from the graph to tick (✓) the **three** correct statements. [3]

Every five years the required thickness of loft insulation increases.	☐
The required thickness of loft insulation in 2000 is 8 times thicker than in 1970.	☐
In 1960 houses did not lose any energy through their roof.	☐
A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards.	☐
The **general** trend of the graph indicates that the required thickness of loft insulation has increased at an increasing rate.	☐
The required thickness of loft insulation will remain constant after 2015.	☐

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(iii) A homeowner must install loft insulation in a new extension. **It has a loft area of 120 m^2**. The insulation must be at least 270 mm thick to meet building regulations.

There is a selection of fibre-glass insulations available.

	Insulation 1 (270 mm thick)	Insulation 2 (350 mm thick)	Insulation 3 (300 mm thick)
Installation cost (£/m^2)	3.50	5.55	4.50
Estimated saving per year (£)	84	111	98
Payback time (years)	5.0	6.0	

Calculate the payback time if **insulation 3** was installed in the 120 m^2 extension. [2]

Payback time = ... years

(iv) The homeowner considers installing **insulation 1** as it is cheapest but the builder says that **insulation 2** should be installed as it will save more money over 40 years. Explain, with calculations, whether the builder is correct. [2]

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